

Jake Hoffman

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LinkedIn

jakethehoffer.github.io/website

SUMMARY OF QUALIFICATIONS

- Computer Engineering student focused on software, automation, and systems that run unattended.
- Proficient in Python, Java, C, C#, C++, SQL, Git, Word, Excel, JavaScript, Dart, Unity, FastAPI, Playwright, Monday.com, and Google Apps Script.

EDUCATION

4th Year at Queen's University

September 2023 – Present

Computer Engineering · Expected graduation 2027

- Cumulative GPA of 3.85 with distinction of Dean's Scholar.

EXPERIENCE

Operations Technology Intern · Digital Operations Improvement

Jun 2026 – Aug 2026

Lynx Equity Limited (3rd consecutive summer)

- Shipped releases on the company's investor website through the Webflow API, each gated on scripted checks against staging and production. A round of 14 changes went from request to live in an afternoon, against two days by hand.
- Replaced all 51 portfolio-company bios in one scripted API pass, checked by a rule built to break if a bio landed on the wrong company. All 51 mapped correctly, with 3 extraction defects caught before publish.
- Verified 7 stage transitions across the 6 boards of the portfolio's Monday.com system, shipped a 5-field gate enforced in the UI and API, and traced a defect measuring project age from approval rather than start.

Independent Software Engineer · Trading & Automation Systems

Nov 2025 – Present

- Operate an S&P 500 paper-trading agent and an arbitrage daemon across nineteen bookmakers, both 24/7 in production since April 2026 and November 2025 (detailed under Personal Projects).
- Shipped market-bot, a crypto market-making feasibility study, in three weeks. 7 days of 24/7 venue telemetry produced a NOT_READY verdict across all 75 pairs.

IT Intern

Jun 2024 – Aug 2024; Jun 2025 – Aug 2025

Lynx Equity Limited

- Built a task management system on Monday.com that routed issues to the right workflow queue across subsidiaries and internal departments.
- Wrote Google Apps Script (JavaScript) that consolidates multiple Excel exports into one Google Sheet, applies pricing logic, and tracks unmatched keys for reconciliation.

Game Developer (Independent)

January 2024 – March 2024

Mega Tic-Tac-Toe

- Built a Unity game on a 3×3 grid of linked Tic-Tac-Toe boards where each move sets the opponent's next board, and winning a board claims one square. Supports local and online multiplayer, plus single player vs. a Minimax AI.

Design Team Member

October 2023 – March 2024

QMIND · Queen's AI Hub

- Trained a Blackjack agent with reinforcement learning on a team project. Built a tabular Q-learning baseline with training loop and tests, then helped migrate it to Deep Q-learning so the model could learn policies beyond a lookup table.

PERSONAL PROJECTS

trader (Python · IBKR · Finnhub · pytest)

- 24/7 AI swing-trading agent for S&P 500 equities, driven by six scheduled, idempotent routines.

Odds Aggregator (Python · Playwright · SQLite/Alembic · FastAPI)

- Production arbitrage daemon covering 19 bookmakers across 7 sports.

tax-rebalance (Python · async SQLAlchemy 2 · httpx · pytest)

- Canadian TFSA + RRSP portfolio drift monitor that emails weekly digests with cost-aware rebalance verdicts (read-only; never places trades). Spec-driven TDD: 21-task plan, two-stage independent review, 117 tests. Questrade IQ API with OAuth refresh-token rotation.

equity-arbs (Python · asyncio · SQLAlchemy 2 · statsmodels · pydantic)

- Stat-arb research toolkit for TSX pair trading (spec-first, TDD, 76 unit tests). Backtest demonstrated a ~0.6%/yr ceiling on retail capital; retired at v1.0 as a documented negative result.

Writing: jakethehoffer.github.io/website/#writing, short essays on engineering tradeoffs.